

Cognizant Deepskilling Modules

Engineering Concepts:

DSA Exercises:

Exercise 1: Inventory Management System – Output

```
Current Inventory:
Product ID: 101, Name: Laptop, Quantity: 10, Price: ₹55000.0
Product ID: 102, Name: Mouse, Quantity: 50, Price: ₹600.0
Product ID: 103, Name: Keyboard, Quantity: 20, Price: ₹1200.0

Updating Product...
Product Updated Successfully.

Inventory After Update:
Product ID: 101, Name: Laptop, Quantity: 15, Price: ₹53000.0
Product ID: 102, Name: Mouse, Quantity: 50, Price: ₹600.0
Product ID: 103, Name: Keyboard, Quantity: 20, Price: ₹1200.0

Deleting Product...
Product Deleted Successfully.

Final Inventory:
Product ID: 101, Name: Laptop, Quantity: 15, Price: ₹53000.0
Product ID: 103, Name: Keyboard, Quantity: 20, Price: ₹1200.0

Process finished with exit code 0
```

Exercise 2: E-commerce Search – Output

```
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Users\msree\AppData\Local\Programs\IntelliJ IDEA 2026.1.4\lib\idea_rt.jar=60758" -Dfile.encoding=UTF-8
Linear Search:
ID: 103, Name: Keyboard, Category: Accessories

Binary Search:
ID: 103, Name: Keyboard, Category: Accessories

Process finished with exit code 0
```

Exercise 3: Sorting Customer Orders – Output

```
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Users\msree\AppData\Local\Programs\IntelliJ IDEA 2026.1.4\lib\idea_rt.jar=60758" -Dfile.encoding=UTF-8
Original Orders:
Order ID: 101, Customer: Rahul, Total Price: ₹2500.0
Order ID: 102, Customer: Priya, Total Price: ₹5000.0
Order ID: 103, Customer: Amit, Total Price: ₹1200.0
Order ID: 104, Customer: Sneha, Total Price: ₹4000.0

After Bubble Sort:
Order ID: 103, Customer: Amit, Total Price: ₹1200.0
Order ID: 101, Customer: Rahul, Total Price: ₹2500.0
Order ID: 104, Customer: Sneha, Total Price: ₹4000.0
Order ID: 102, Customer: Priya, Total Price: ₹5000.0

After Quick Sort:
Order ID: 103, Customer: Amit, Total Price: ₹1200.0
Order ID: 101, Customer: Rahul, Total Price: ₹2500.0
Order ID: 104, Customer: Sneha, Total Price: ₹4000.0
Order ID: 102, Customer: Priya, Total Price: ₹5000.0
```

Exercise 4: Employee Management System – Output

```

"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Users\msree\AppData\Local\Programs\IntelliJ IDEA 2026.1.4\lib\idea_rt.jar
Employee Added Successfully.
Employee Added Successfully.
Employee Added Successfully.

Employee List:
Employee ID: 101, Name: Rahul, Position: Manager, Salary: ₹60000.0
Employee ID: 102, Name: Priya, Position: Developer, Salary: ₹50000.0
Employee ID: 103, Name: Amit, Position: Tester, Salary: ₹45000.0

Searching Employee:
Employee ID: 102, Name: Priya, Position: Developer, Salary: ₹50000.0

Deleting Employee:
Employee Deleted Successfully.

Employee List After Deletion:
Employee ID: 101, Name: Rahul, Position: Manager, Salary: ₹60000.0
Employee ID: 103, Name: Amit, Position: Tester, Salary: ₹45000.0

Process finished with exit code 0

```

Exercise 5: Task Management System – Output

```

Task Added Successfully.
Task Added Successfully.
Task Added Successfully.

Task List:
Task ID: 101, Task Name: Design UI, Status: Pending
Task ID: 102, Task Name: Develop Backend, Status: In Progress
Task ID: 103, Task Name: Testing, Status: Pending

Searching Task:
Task ID: 102, Task Name: Develop Backend, Status: In Progress

Deleting Task:
Task Deleted Successfully.

Task List After Deletion:
Task ID: 101, Task Name: Design UI, Status: Pending
Task ID: 103, Task Name: Testing, Status: Pending

Process finished with exit code 0

```

Exercise 6: Library Management System – Output

```

"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Users\msree\AppData\Local\Programs\IntelliJ IDEA 2026.1.4\lib\idea_rt.jar=63610" -Dfile.encoding=l
Linear Search:
Book ID: 104, Title: Java Programming, Author: Herbert Schildt

Binary Search:
Book ID: 104, Title: Java Programming, Author: Herbert Schildt

Process finished with exit code 0

```

Exercise 7: Financial Forecasting – Output

```

"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Users\msree\AppData\Local\Programs\IntelliJ IDEA 2026.1.4\lib\idea_rt.jar=63606" -Dfile.encoding=l
Present Value : ₹10000.0
Growth Rate   : 10.0%
Years         : 5
Future Value  : ₹16105.10

Process finished with exit code 0

```

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